

Partial and Multiple correlation:-

It is often important to measure the correlation between a dependent variable and one particular independent variable, when all other variables involved are kept constant i.e. when the effect of all other variables are removed.

This can be obtained by calculating coefficient of partial correlation.

for ex:- if we have three variables - yield of wheat, amount of rainfall and temperature and if we limit our analysis to yield and rainfall, when a certain daily temp existed. or if we treat the problem mathematically in such a way that changes in temperature are not allowed. This problem becomes one of partial correlation.

i.e.

partial correlation measures the degree of relationship between y and one of the variables $x_1, x_2, \dots, x_n$, with the effect of all other variables removed.

If we denote r_{12} , the coeff. of partial correlation between x_1 and x_2 , keeping x_3 constant we find that

$$r_{12,3} = \frac{r_{12} - r_{13} r_{23}}{\sqrt{1 - r_{13}^2} \sqrt{1 - r_{23}^2}}$$

slly coeff of partial correlation b/w x_1 & x_3
keeping x_2 constant

$$r_{13.2} = \frac{r_{13} - r_{12} \cdot r_{23}}{\sqrt{1 - r_{12}^2} \sqrt{1 - r_{23}^2}}$$

slly partial correlation coeff b/w x_2 & x_3
keeping x_1 constant

$$r_{23.1} = \frac{r_{23} - r_{12} r_{13}}{\sqrt{1 - r_{12}^2} \sqrt{1 - r_{13}^2}}$$

It should be noted that value of a partial correlation coefficient is always interpreted, by squaring the partial correlation coefficient.

Thus if x_1 , x_2 , x_3 ~~rep~~ represent sales, advertisement expenditure and price respectively and we get $r_{12.3}^2 = 0.912$

It means that more than 91 percent of variation in x_1 (sales) is associated with x_2 (advertisement expenditure), & not associated with x_3 (price)

Zero, first & second order coefficient:-

$r_{12.3}$, $r_{23.1}$, $r_{13.2}$ are called first order coefficient, since only one variable has been held ~~efficient~~ constant. Simple coeff (b/w two var) called zero order coeff, since no variable held constant.

$r_{12.34}$, $r_{13.24}$, $r_{14.23}$ etc are called second order coefficients since two variable

are kept constant.

$$r_{12.34} = \frac{r_{12.4} - r_{13.4} r_{23.4}}{\sqrt{1 - r_{13.4}^2} \sqrt{1 - r_{23.4}^2}}$$

$$\text{or } r_{12.34} = \frac{r_{12.3} - r_{14.3} r_{24.3}}{\sqrt{1 - r_{14.3}^2} \sqrt{1 - r_{24.3}^2}}$$

Similarly

$$r_{13.24} = \frac{r_{13.2} - r_{14.2} r_{34.2}}{\sqrt{1 - r_{14.2}^2} \sqrt{1 - r_{34.2}^2}}$$

$$\text{or } r_{13.24} = \frac{r_{13.4} - r_{12.4} r_{23.4}}{\sqrt{1 - r_{12.4}^2} \sqrt{1 - r_{23.4}^2}}$$

Multiple correlation coefficient:-

Multiple correlation measures the degree of relationship b/w dependent variable say x_1 and other variables $x_2, x_3, x_4, \dots$. The variable whose value we are trying to estimate is called dependent variable and other variables on which our estimation is based are known as independent variable.

In multiple correlation, we always have three or more variables. The dep. var. is denoted by X_1 & other by $X_2, X_3, X_4, \dots$ etc.
coeff. of multiple correlation :- It is denoted

by R_1 and it is common to add subscript designating the variables involved.

Thus $R_{1.234}$ would represent the coeff of multiple correlation b/w X_1 on one hand and X_2, X_3, X_4 on the other.

Subscript of dep. var. is always to the left of the point.

$$R_{1.23} = \sqrt{\frac{\sigma_{12}^2 + \sigma_{13}^2 - 2\sigma_{12}\sigma_{13}\sigma_{23}}{(1 - \sigma_{23}^2)}}$$

$$R_{2.13} = \sqrt{\frac{\sigma_{12}^2 + \sigma_{23}^2 - 2\sigma_{12}\sigma_{13}\sigma_{23}}{(1 - \sigma_{13}^2)}}$$

slly

$$R_{3.12} = \sqrt{\frac{\sigma_{13}^2 + \sigma_{23}^2 - 2\sigma_{12}\sigma_{13}\sigma_{23}}{(1 - \sigma_{12}^2)}}$$

$$0 \leq R_{1.23} \leq 1$$

ie it lies b/w 0 and 1. It is closer to 1, the better is the linear relationship b/w the variables. The closer to

0, the worse is the linear relationship. if it is 1, perfect correlation, 0 mean no relationship.

note that :- correlation coeff lies b/w -1 to 1 but the multiple correlation coeff. is always +ve and lies b/w 0 & 1.

Q:- If $r_{12} = 0.86$, $r_{13} = 0.65$, $r_{23} = 0.72$
find the partial correlation coefficient
 $r_{12.3}$.

Soln:-

$$r_{12.3} = \frac{r_{12} - r_{13} r_{23}}{\sqrt{1 - r_{13}^2} \sqrt{1 - r_{23}^2}}$$

$$= \frac{.86 - .65 \times .72}{\sqrt{1 - (.65)^2} \sqrt{1 - (.72)^2}}$$

$$[r_{12.3} = 0.743]$$

Q Is It possible to get the following
from a set of experiental data. the value
of $r_{12.3}$.

(a) $r_{23} = 0.8$, $r_{13} = -0.5$, $r_{12} = 0.6$

(b) $r_{23} = 0.7$, $r_{31} = -0.4$, $r_{12} = 0.6$

Soln:-

$$r_{12.3} = \frac{r_{12} - r_{13} r_{23}}{\sqrt{1 - r_{13}^2} \sqrt{1 - r_{23}^2}}$$

$$r_{12.3} = 1.923$$

Since $r_{12.3}$ is greater than one, there
is some inconsistency in the given data.

(b) $r_{12.3} = 1.344$

again greater than 1, which is not possible.

ex:- The following zero order correlation coefficients are given.

$$r_{12} = .98 \quad r_{13} = 0.44 \quad r_{23} = 0.54$$

calculate multiple correlation coefficient treating first variable as dependent and second and other as independent.

$$R_{1.23} = \sqrt{\frac{r_{12}^2 + r_{13}^2 - 2r_{12}r_{13}r_{23}}{(1 - r_{23}^2)}}$$

$$[R_{1.23} = .986]$$

Q From the following table, obtain $R_{1.23}$, $R_{2.13}$, $R_{3.12}$

$$X_1 : 2 \quad 5 \quad 7 \quad 11$$

$$X_2 : 3 \quad 6 \quad 10 \quad 12$$

$$X_3 : 1 \quad 3 \quad 6 \quad 10$$

soln we need r_{12} , r_{13} , r_{23}

x_1	x_2	x_3	x_1^2	x_2^2	x_3^2	$x_1 x_2$	$x_2 x_3$	$x_1 x_3$
2	3	1	4	9	1	6	2	3
5	6	3	25	36	9	30	15	15
7	10	6	49	100	36	70	42	42
11	12	10	121	144	100	132	110	110
25	31	20	199	289	146	238	169	122

$$r_{12} = \frac{N \sum x_1 x_2 - \sum x_1 \sum x_2}{\sqrt{N \sum x_1^2 - (\sum x_1)^2} \sqrt{N \sum x_2^2 - (\sum x_2)^2}}$$

$$r_{12} = \frac{4 \times 146 - 25 \times 31}{\sqrt{4 \times 199 - (25)^2} \sqrt{4 \times 289 - (31)^2}}$$

$$r_{12} = 0.97$$

$$\text{Similarly } r_{13} = 0.99$$

$$r_{23} = 0.97$$

Now

$$R_{1.23} = \sqrt{\frac{r_{12}^2 + r_{13}^2 - 2 r_{12} r_{23} r_{13}}{1 - r_{23}^2}}$$

$$R_{1.23} = 0.99$$

$$R_{2.13} = \sqrt{\frac{r_{12}^2 + r_{23}^2 - 2 r_{12} r_{23} r_{13}}{1 - r_{13}^2}}$$

$$R_{2.13} = 0.97$$

$$R_{3.12} = 0.99$$